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ORIGINAL PAPER

Effects of Social Stories on Increasing Social Interaction and Engagement of Deaf and Hard of Hearing Students with Autism Spectrum Disorder in Inclusive Settings

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Abstract
d/Deaf or hard of hearing (d/Dhh) students with autism spectrum disorder (ASD) may require specific approaches to promote social inclusion. The purpose of this study was to investigate the effects of reading social stories with American Sign Language vocabulary to students who are d/Dhh with ASD (d/Dhh-ASD) and their peers in inclusive settings, using a non-concurrent multiple baseline design across participants. To examine the effectiveness of the intervention, the frequency of four communicative social behaviors and the duration of social engagement with peers were recorded for each participant during lunchtime and recess. The results did not show an immediate effect across all participants. Implications for promoting social inclusion for students who are d/Dhh-ASD in general education classrooms are discussed.

Keywords Social stories · Deaf · Hard of hearing · Autism spectrum disorder · Socialization · Inclusion

Introduction

Evidence-based practices (EBP) to increase communication and social interaction of students who are deaf and hard of hearing (d/Dhh) with autism spectrum disorder (ASD) are critically needed, especially when these students are in inclusive settings. Over the past two decades, it has become common practice to teach students who are d/Dhh in general education classrooms with support from an itinerant teacher of the deaf and hard of hearing (TDHH). To successfully include d/Dhh students in inclusive settings, adaptations may be necessary (Raver et al., 2014). Merely placing d/Dhh students in an inclusive setting may not automatically facilitate meaningful social interaction, peer acceptance, and/or improvement in the students' social communication skills (Antia et al., 1993, 2002; Bobzien et al., 2013; Hyde & Power, 2004; Weisel et al., 2005). Approximately 40% of d/Dhh students present with one or more disabilities (Gallaudet Research Institute [GRI], 2013) including ASD. One in every 59 d/Dhh students is estimated to exhibit characteristics of ASD (Szymanski et al., 2012). Considering that students with these additional challenges often require more specific approaches to promote social integration, there is a dearth of research on practices that can guide practitioners in promoting social interaction and communication between students who are d/Dhh with ASD (d/Dhh-ASD) and their peers in general education classrooms.

Social Interactions Between Peers

Peer interaction is a fundamental component of social development and necessary for all children. Educators in public schools are often concerned about socially integrating children with varying levels of hearing since they may not easily participate in social activities with their peers due to limited access to spoken language and verbal cues (Bat-Chava & Deignan, 2001; DeLuzio & Girolametto, 2011; Peterson, 2004; Stinson & Antia, 1999). Given that children with ASD may have deficits in social communication and interaction (American Psychiatric Association, 2013), challenges may be much greater in children who have both a hearing loss and an ASD (Davis et al., 2010). Researchers suggest modifying established practices from the field of ASD or

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multiple disabilities because (a) little guidance is provided regarding effective interventions related to communication and social skills for children who are d/Dhh-ASD (Bruce & Borders, 2015) and (b) interventions from the field of d/Dhh itself may be inadequate to meet the needs of this complex population (Borders et al., 2016). Professionals providing services to learners are urged to look to adjacent fields when determining intervention practices, specifically those from the field of ASD and severe and multiple disabilities which have been developed to address communication, socialization, and behavioral needs (Borders et al., 2022).

Social Stories as an Evidence-Based Practice

Social stories or social narratives is one of the 27 interventions that the National Professional Development Center on Autism Spectrum Disorder identified as EBPs for students with ASD (Wong et al., 2015). A social story describes a situation or concept in explicit terms, depicting relevant social cues, perspectives, and common responses of typically-developing individuals (Gray, 2004). This intervention has been shown to be an effective strategy in improving social interaction and communication in children with ASD (Chan & O'Reilly, 2008; Crozier & Tincani, 2007; Schneider & Goldstein, 2010), as well as in improving the classroom behavior of students with intellectual disabilities (Beh-Pajooh et al., 2011; Kim et al., 2014).

Modifications for d/Dhh Students with ASD

To use social story interventions for students who are d/Dhh-ASD, modifications might be necessary. In previous studies, social stories were introduced as spoken or written language with or without visuals (Raver et al., 2014). However, in a meta-analysis of social story interventions for students with ASD, Kokina and Kern (2010) found that if a child has difficulty reading or understanding spoken language, using written text as a method of explaining social stories is ineffective. When using any of the EBPs established in the field of ASD for students who are d/Dhh-ASD, it is important to consider potential modifications to the intervention (Borders et al., 2016, 2022; Bruce & Borders, 2015). Modifications suggested in the literature to suit the unique needs of learners who are d/Dhh-ASD include: (a) inclusion of their peers in implementation of the intervention and (b) incorporation of American Sign Language (ASL) vocabulary to increase communication between learners who are d/Dhh-ASD and their peers.

First, both learners who are d/Dhh-ASD and their peers should be included when implementing an intervention involving social interactions and incorporating communication (Borders et al., 2022). Fees et al. (2014) introduced a social story to all of the students in six inclusive preschool classrooms instead of teaching only the targeted students with diverse developmental disabilities. The researchers reported that the intervention improved both individual and group behaviors, including students with disabilities and those without. When peers were engaged, they regulated their own behavior and also served as models or assistants to teachers for children with developmental challenges (Fees et al., 2014). Unlike academic or literacy skills, social or communicative behaviors involve peers as communication partners; thus, it is essential to include both the learner who is d/Dhh-ASD and peers in social skill interventions to effectively change behaviors.

Second, when implementing social skill intervention for students who are d/Dhh-ASD, it is important to include strategies for incorporating communication that can be used by both the student who is d/Dhh-ASD and their peers (Borders et al., 2022). When students who are d/Dhh-ASD in inclusive settings use ASL, social interactions with peers may become more challenging (Borders et al., 2022) because teachers, educational assistants, and peers typically have minimal to no ASL knowledge or skills. Classroom peers may not have language tools to effectively interact with the d/Dhh students who communicate using ASL. This may cause them to withdraw and remain as spectators. Thus, when implementing social stories for learners who are d/Dhh-ASD, teaching ASL vocabulary related to the social story to all of the children in a classroom may increase social interactions between students who are d/Dhh-ASD and their peers.

Despite the unique aspects of teaching learners who are d/Dhh-ASD, only one empirical study to date has examined the effect of social stories for d/Dhh students in inclusive settings. Raver et al. (2014) evaluated the effects of dyad-specific social stories on the communication and social skills of four preschoolers with varying hearing levels in their inclusive preschools. Four children with typical hearing levels from the inclusive preschools served as play-partners for the target participants with hearing loss. Results showed improvements in verbal comments and play turns from three of the four d/Dhh participants. When interventions that are effective for learners with ASD are used, the intervention needs to be modified in consideration of the unique communication aspects d/Dhh learners may experience (e.g., limited access to auditory signals or use of signed systems of communication; Borders et al., 2016).

Research Questions

The purpose of this study is to investigate the effectiveness of an intervention to increase social interactions and communicative engagement between students who are d/Dhh-ASD and their peers in inclusive settings. Specifically, the research questions examine whether a functional relation

exists between social stories and improvements in social interaction among students who are d/Dhh-ASD and their peers in inclusive settings. The social stories for this study were modified to suit the needs of students who are d/Dhh-ASD (Borders et al, 2016; Bruce & Borders, 2015), including (a) teaching the stories to the whole class, not only to the d/Dhh student with ASD; and (b) the addition of ASL vocabulary.

We address the following research questions:

1. Does whole-class implementation of social stories with ASL vocabulary increase (a) social initiations and responses of d/Dhh students with ASD, (b) social initiations and responses of their peers, and (c) social engagement between d/Dhh students with ASD and their peers?
2. Do the educators perceive the goals, procedures, and outcomes of the intervention as socially important and acceptable in their inclusive settings?

Method

Participants

Three d/Dhh students with ASD, along with 63 classroom peers in three different general education classrooms (with approximately 21 students in each class) agreed to participate in this study. Inclusionary criteria consisted of d/Dhh students who: (a) use ASL; (b) have been diagnosed with ASD, as confirmed by their teachers and parents; (c) are in primary school; (d) have limited social interaction skills with their peers at school, as reflected in teachers' reports and researcher observations; and (e) are the only student who is d/Dhh-ASD in their general education class. Table 1 shows the demographic characteristics of the three d/Dhh participants. To ensure confidentiality, all participants were given pseudonyms. All three participants attended a third grade general education classroom with full-time Educational Assistant (EA) support. The skill sets of each EA varied, and the level of peers' exposure to ASL in the school setting also varied.

Measures

To examine the effectiveness of the intervention, the frequency of four communicative social behaviors and the duration of social engagement with peers were observed. Because the d/Dhh students used ASL, words, signs, and gestures, these aspects of communication were coded equivalently. Interactions toward any adult or in response to an adult's prompt were not coded. Definitions, examples, and non-examples for the observed behaviors are detailed in

Table 1 Demographics of d/Dhh participants

Participants	Hearing status	Disability(ies)	Language and literacy	Instructional setting
Jimin	Profound hearing level; inner ear anomalies bilaterally; no assistive technology	Diagnosed with ASD at age of 3; moderate intellectual delays	Primary language is ASL; emerging one and two-word phrases (e.g., red sock, more swimming), no literacy skills yet	Support from full-time EA with behaviour intervention experience and ASL skills; 22 peers in the class were exposed to ASL daily on the playground and in the hallway where students and staff from the School for the Deaf shared their school setting
Ali	Moderate hearing level; uses hearing aids, personal FM system, and soundfield system at school	Diagnosed with ASD at an early age	Uses three to four-word ASL sentences (e.g., (you) want (to) play (with) me?); uses some spoken Arabic at home; emerging level English language and literacy	Support from full-time EA and ASL interpreter; 20 peers in the class were exposed to ASL interpreters, Deaf EA, and Deaf Language Acquisition Support Workers in their school setting
Rain	Severe to profound hearing level; no assistive technology used	Diagnosed with ASD at an early age, cerebral palsy, developmental delays due to Cytomegalovirus during pregnancy	Uses one-word ASL signs (e.g., eat, more) and writes one to two-word phrases; comprehends simple written messages (e.g., Time to work now, iPad off, wait, Math time), some behavioral issues (i.e., grabbing, poking others)	Support from full-time EA with some ASL skills; 21 peers in the class were not exposed to ASL in the school prior to student enrolling in class

Table 2 Coding definitions, examples, and non-examples

Code	Definition	Examples	Non-examples
Social initiation: Initiation by d/Dhh student; Initiation by peer	Verbal, signing, or gestural initiations that are directed toward a peer in neutral or positive manner	Greeting; praising; questioning; commenting on the peer's activity; handing a toy to another peer; asking if the peer wishes to join an activity; offering assistance during play; holding their hand and moving together toward	Any interaction directed to an adult or by an adult's prompt; any negative verbal or nonverbal interaction; unintelligible utterance that is not directed to any peer; self-stimulation without a peer within a meter (e.g., oops, oh no, yeah)
Social response: Response from d/Dhh student; Response from peer	Verbal, signing, or gestural response toward a peer who has initiated positive/ neutral social behavior; and each consecutive social interaction after the first initiation	Complying with another student's request; replying to a verbal or signing interaction; looking at a child in response to their name sign or to shoulder-tapping by a peer	A behavior that occurs no later than 10 s after a peer's initiation or response
Social engagement with peer	Saying one or more understandable words or signs while positioned within 3–5 ft of a peer and with body oriented toward the peer, directing a gesture/ movement toward a peer, making a cooperative response, playing with toys or materials as intended, or imitating a peer	Playing chase; adding a block to a structure peers are building together; sharing materials playing a cooperative game; running to the monkey bars with a peer	Wandering; sitting on a swing or at the top of the slide alone; walking in a repetitive sequence; leaves the activity area (<3 to 5 ft outside of specified area); inappropriate problem behaviors (e.g., destroying or throwing item toward a peer); staying with group but parallel play

The coding scheme used for this study has been adapted from the prior work of Delano and Snell (2006), Ledford et al., (2016), Owen-DeSchryver et al., (2008), and others using similar codes for social interaction research (e.g., Davis et al., 1996; Thiemann & Goldstein, 2001)

Table 2 and discussed below. Using a smartphone software application (i.e., Behavioral Timer) all social interactions were measured using a direct observation coding system (Ledford et al., 2016). For all observations, data were collected for 20 min: 10 min in the classroom during lunchtime and another 10 min on the playground during recess time following lunch.

Communicative Social Behaviors

The frequencies of four communicative social behaviors were measured using a frequency or event recording system: (a) d/Dhh student's social initiation towards a peer, (b) peer's social initiation toward the d/Dhh student, (c) d/Dhh student's response to the peer's initiation, and (d) peer's response to the d/Dhh student's initiation. The number of social initiations and responses made by the d/Dhh students and their peers were recorded in each 20-min session. When a conversation occurred between a d/Dhh student and a peer, each consecutive social interaction after the first initiation was scored as a response (Owen-Deschryver et al., 2008).

Social Engagement

Duration of social engagement with peers included the time that the d/Dhh student spent interacting with their peers or engaged appropriately in activities in proximity with peers. Playing with materials or using facilities on the playground appropriately, however, did not count as "engaged" if the d/Dhh student or peer was by themselves or not interacting with a peer during the time observed. Duration of social engagement between the d/Dhh student and his/her peers was calculated using a 10-s momentary time sampling procedure (Ledford et al., 2016). At the end of each 10-s interval, cued by a tone or vibration from their phone, data collectors coded either "engaged (occurrence)" or "not engaged (non-occurrence)." For each session, the researcher calculated the percentage of intervals that the d/Dhh student was engaged with peers to determine the time spent together. For example, if there were 20 intervals coded as "engaged" for 10 min of an observation, the percentage of intervals with engagement would be 33% (total number of 10-s intervals for 10 min is 60; $20/60 \times 100 = 33$).

Social Validity

Classroom teachers and educational assistants completed a questionnaire to rate statements regarding the social validity of the intervention on a Likert-type scale from 1 (strongly disagree) to 5 (strongly agree). The statements included

the importance of skills taught, perceived effectiveness, appropriateness, and future use in their classrooms (Chan & O'Reilly, 2008).

Procedures

Recruitment and Consent

After obtaining university ethics and school district research approval, school principals and classroom teachers were contacted to see if they were interested in participating in this study. Those who were interested then sent home an explanation of the study and parent consent forms to potential d/Dhh student participants. The first three available participants who met the inclusionary criteria and whose parents consented were included as participants for the study.

Research Design and Setting

Single case research methodology, also referred to as single-subject experimental research, was used in this study to assess the effects of social stories with ASL vocabulary on social interactions between students who are d/Dhh-ASD and their hearing peers in inclusive settings. While case studies are purely descriptive and not experimental, single-case designs can be used to examine the effectiveness of an intervention through a functional relationship between an intervention and its outcomes (Cannon et al., 2016; Kazdin, 2011; Kratochwill et al., 2010). This methodology has been acknowledged for decades as it is well suited to address educational practices and answer applied problems (Barlow et al., 1984; Odom, 1988; Tawney & Gast, 1984), especially with low-incidence populations such as d/Dhh students (Cannon et al., 2016; Guardino & Antia, 2012; Bullis & Anderson, 1986). One of the most common single case designs is multiple baseline design, in which researchers replicate the comparison between the baseline (i.e., pre-intervention condition) and intervention outcomes across at least three different behaviours, contexts, or participants to ensure internal validity (Cannon et al., 2016; Ledford et al., 2018). In this study, however, as the d/Dhh participants were integrated across three different schools in two different school districts, establishing contemporaneous baselines would have been very difficult; therefore, a non-concurrent multiple baseline design across participants was used (Ledford et al., 2018).

The first author observed and collected data on the targeted social behaviors of the students who are d/Dhh-ASD and their peers during the lunchtime and recess periods. The data were collected in the classroom during lunch and on the playground during recess because these unstructured and non-instructional times provide consistent opportunities for students to participate in social interactions (Kurth et al., 2015; More et al., 2013; Owen-Deschryver et al., 2008). When conducting observations, the first author remained as far away from the students who are d/Dhh-ASD as possible while still ensuring that they could hear any verbal interactions or see signs and gestures. This study had three phases: baseline, intervention, and follow-up.

During the baseline phase, data collection was conducted three days per week until a stable level and trend occurred, at which point the intervention was introduced to the first participant who was d/Dhh-ASD and their peers. During baseline, the classroom teachers read stories to the whole class or provided students with free time to complete independent work. This process was repeated for all remaining students.

Each intervention session was comprised of reading a social story and teaching five targeted ASL vocabulary, which were embedded in the story. The first author modeled ASL vocabulary as she read each social story. A total of six social stories were used in this study in each setting. The researcher created the stories following Gray's (2004) guidelines. The stories were reviewed by two professors in Special Education at a local university who are experts in d/Dhh and ASD respectively and two elementary school teachers to ensure their compliance with Gray's criteria and appropriateness of the content. The stories were intended to provide the students with prosocial communication strategies in order to increase social interactions between students who are d/Dhh-ASD and their peers in inclusive settings. Topics were specific to the activities during lunchtime and recess, including greetings, expressing feelings, sharing and turn-taking, and playing and working together. Each story consisted of six to seven PowerPoint slides and contained visual illustrations directly related to the text, based on recommendations from previous research (Kokina & Kern, 2010).

The intervention sessions took place just before lunch since social stories are recommended to be read immediately prior to the targeted activities in previous literature (Kim et al., 2014). The classrooms were also used as lunchrooms, so students remained in the same room after the intervention. The 15- to 20-min intervention sessions were conducted three times per week by the researcher for two weeks.

The first follow-up observation occurred two weeks after the final intervention session for each participant. They were conducted three times, once every two weeks. The classroom teachers and educational assistants were asked to answer the social validity questionnaire for the second time, four weeks after the intervention ceased.

Implementation Fidelity

To ensure that intervention was delivered as designed, an independent observer completed an implementation fidelity

checklist that specified the components of a treatment session in 33% of the intervention sessions across three participants. This variable was calculated by dividing the total number of components completed accurately by the total number of correct plus incorrect components and multiplying by 100 (Maione & Mirenda, 2006).

Interobserver Agreement

The same independent observer also collected data simultaneously on the target behaviors of participants for 15 out of the 43 (35%) observation sessions, balanced across participants and study phases. The first author provided the interobserver with training sessions that included: (a) a review of the coding manual (as shown in Table 2); (b) practice coding by watching sample videos that were taken during a summer community program where children with ASD participated; and (c) pilot observation of actual interactions between the d/Dhh students and peers during lunchtime and recess. Once the interobserver and first author completed the coding practice with 90% or higher accuracy on pre-recorded role plays, pilot observations of the students in the inclusive classrooms started. 100% agreement for two, 20-min, consecutive pilot observations was achieved prior to data collection in the baseline phase. For frequency of social initiations and responses, interobserver agreement (IOA) was calculated by comparing the observers' frequencies of each targeted social behavior within two-minute intervals of the 20-min observation sessions using point-by-point agreement. For duration of social engagement with peers, point-by-point agreement was calculated for each 10-s interval. IOA was calculated by dividing the total number of agreements by the total number of agreements plus disagreements and multiplying by 100. The average IOA was 85% for frequency of social initiations and responses, and the average agreement for duration of social engagement was 99%.

Social Validity

The questionnaire was conducted twice: first at the end of the intervention phase, and second during the follow-up phase. Classroom teachers and educational assistants were asked to evaluate the social significance of goals, procedures, and outcomes of the intervention by answering the social validity questionnaire provided.

Results

The result of the implementation fidelity checklist was 100% in all intervention sessions across all participating classes. As shown in Figs. 1, 2, and 3, the results did not show an immediate effect of the intervention across all participants on increasing social interaction and engagement, although two participants and their peers showed some improvements during the follow-up phase.

Frequency of Social Initiations and Responses of d/Dhh Students

Figure 1 displays the results related to the frequency of social initiations and responses by the participants who were d/Dhh-ASD across the baseline, intervention, and follow-up phases. Each data point represents the number of social initiations and responses during a 20-min observation session.

During the baseline phase, all three participants showed zero to very low, stable levels of unprompted social initiations (ranging from 0 to 1). Jimin had 1 response on average (ranging from 0 to 2). Ali showed 2 responses on average (ranging from 0 to 4). Rain's responses remained zero during this phase.

After the intervention was introduced, Jimin and Rain did not show any changes in the numbers of both social initiations and responses towards their peers. Ali displayed an increase in both social initiations (ranging from 0 to 7) and responses (ranging from 1 to 10) after the first story was introduced to the class. This behavior was sustained for the following three data points; however, both his initiation and response decreased to low levels for the remaining three observations. Both Jimin and Ali showed lower levels of social initiations than social responses. Rain did not demonstrate any attempts for social initiation and responses during the entire intervention phase.

During the follow-up phase each participant showed different trends in the target social behaviors. Jimin showed no social initiations during the follow-up phase, which was consistent with his baseline and intervention phase. The number of his social responses became higher (ranging from 0 to 8) approximately four weeks after the intervention ceased. Social behaviors of Ali, in both initiation (ranging from 1 to 23) and response (ranging from 5 to 12), showed a steep accelerating trend. This trend continued through the final observation, where his initiations were much higher than his responses. During the follow-up phase, Rain remained at zero to one for the number of initiations and responsive social behaviours.

Frequency of Social Initiation and Responses of by Peers

Figure 2 shows the results related to the frequency of social initiations and responses made by peers towards the d/Dhh-ASD student in their class across all three participants. Overall, during the entire three phases across the three classes,

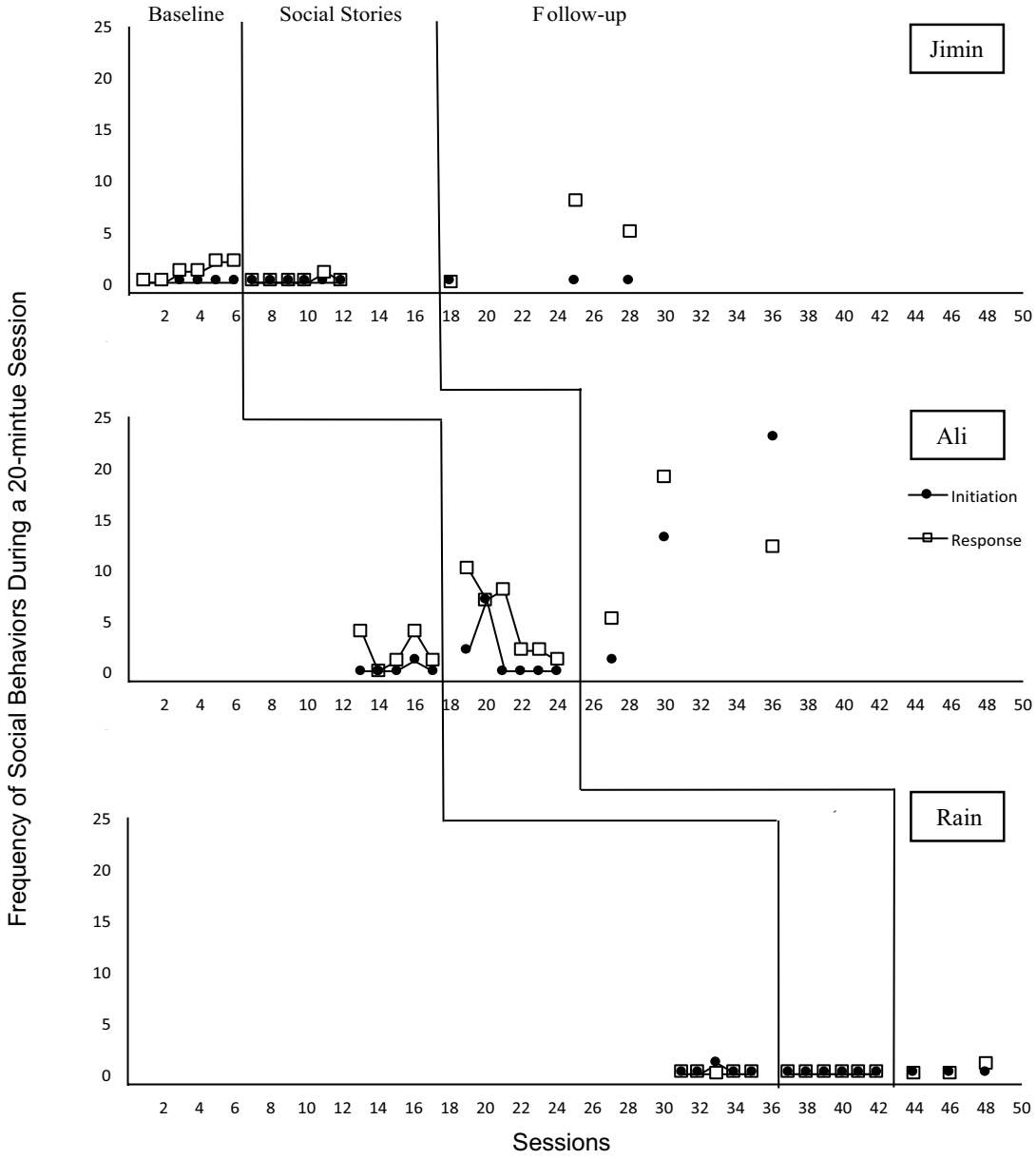


Fig. 1 Frequency of social initiations and responses by d/Dhh students towards their peers

except the follow-up of Ali's peers, peers of the d/Dhh-ASD students initiated (data ranging from 0 to 15) more than they responded (data ranging from 0 to 7).

During the baseline condition, Jimin's peers had a low, stable level of social initiations (ranging from 0 to 2) and responses (ranging from 0 to 1). Ali's peers initially demonstrated 8 social initiations, but they did not initiate during the following session. Baseline data continued to be highly variable throughout the five observation sessions, ranging from 0 to 8. The final data point of baseline showed a decelerating trend of social initiations. Peers' responses to social initiations by Ali were zero to very low and stable, ranging from 0 to 1. Rain's peers initiated and responded one time, each during the baseline phase.

During the intervention phase, social initiation by Jimin's peers were somewhat variable but at a slightly higher level than the baseline (ranging from 0 to 3). There was no social response by Jimin's peers, as no initiation from Jimin occurred. Ali's peers showed an immediate increase in the level of both social initiation and response when the intervention was introduced. However, there was a steep decrease in the middle of the phase, and the social behaviors did not

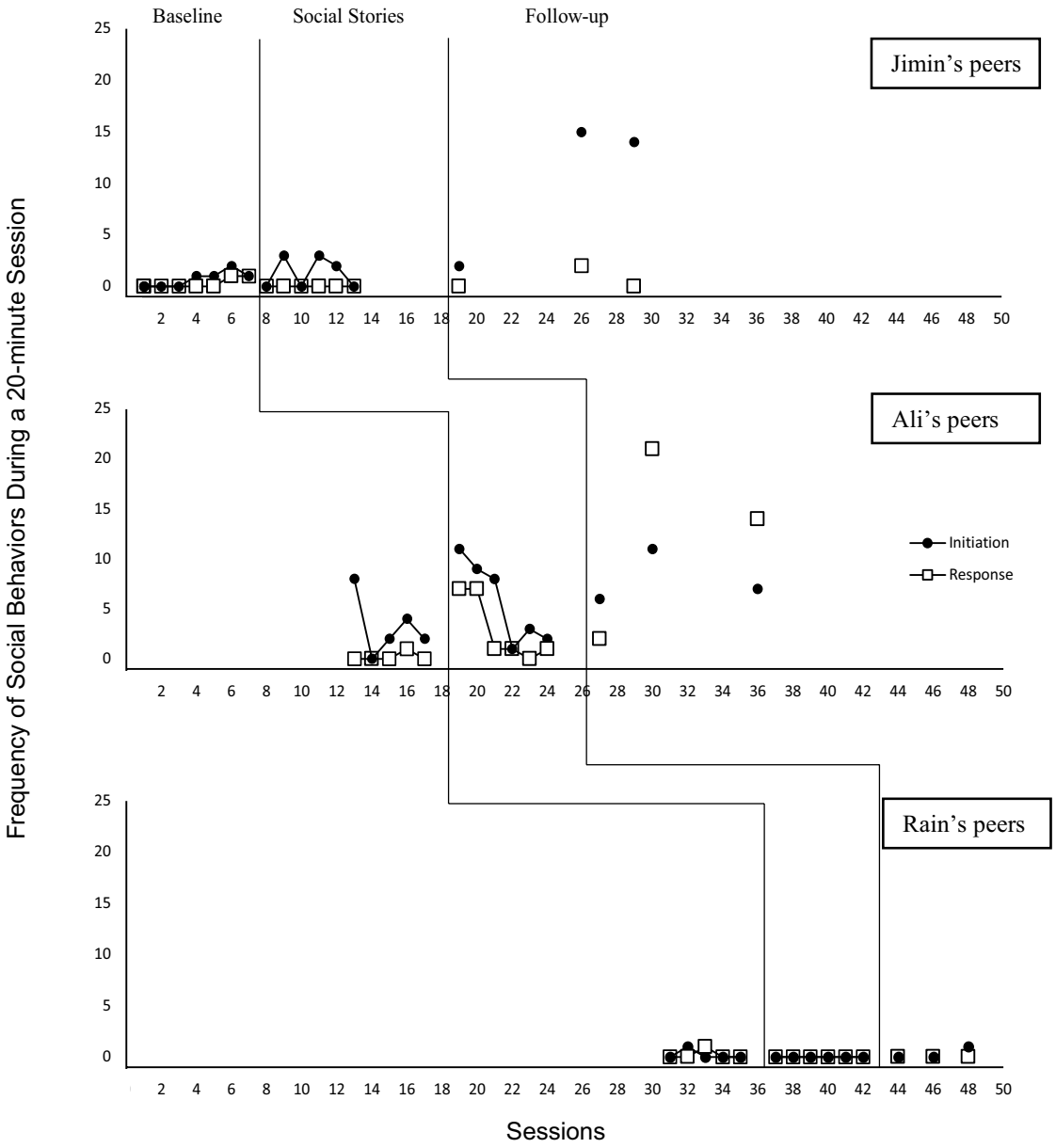


Fig. 2 Frequency of social initiations and responses by peers towards d/Dhh students

recover to the high level seen in the beginning of the intervention phase. There was no communicative attempt made by Rain's peers as the data series is parallel to abscissa.

During the follow-up sessions, Jimin's peers displayed increased initiations across the three observations (ranging from 2 to 15), particularly for the final two data points. The peers responded twice to Jimin's responses during the second observation session of the follow-up phase (ranging from 0 to 2). Ali's peers resumed a similar level of social initiation displayed at the beginning of the intervention phase (ranging from 6 to 11). Their social response data were variable but at a much higher level (ranging from 2 to 14) than the previous phases, aligning with an increase in Ali's initiation. There was a downward trend in both initiating and responding to Ali by the end of the follow-up phase. Peers of Rain attempted one social initiation during the three follow up sessions, which resulted in a response from Rain.

Duration of Social Engagement between the d/Dhh Student and Peers

The graph in Fig. 3 shows the estimated duration that each of the three participants who are d/Dhh-ASD were engaged

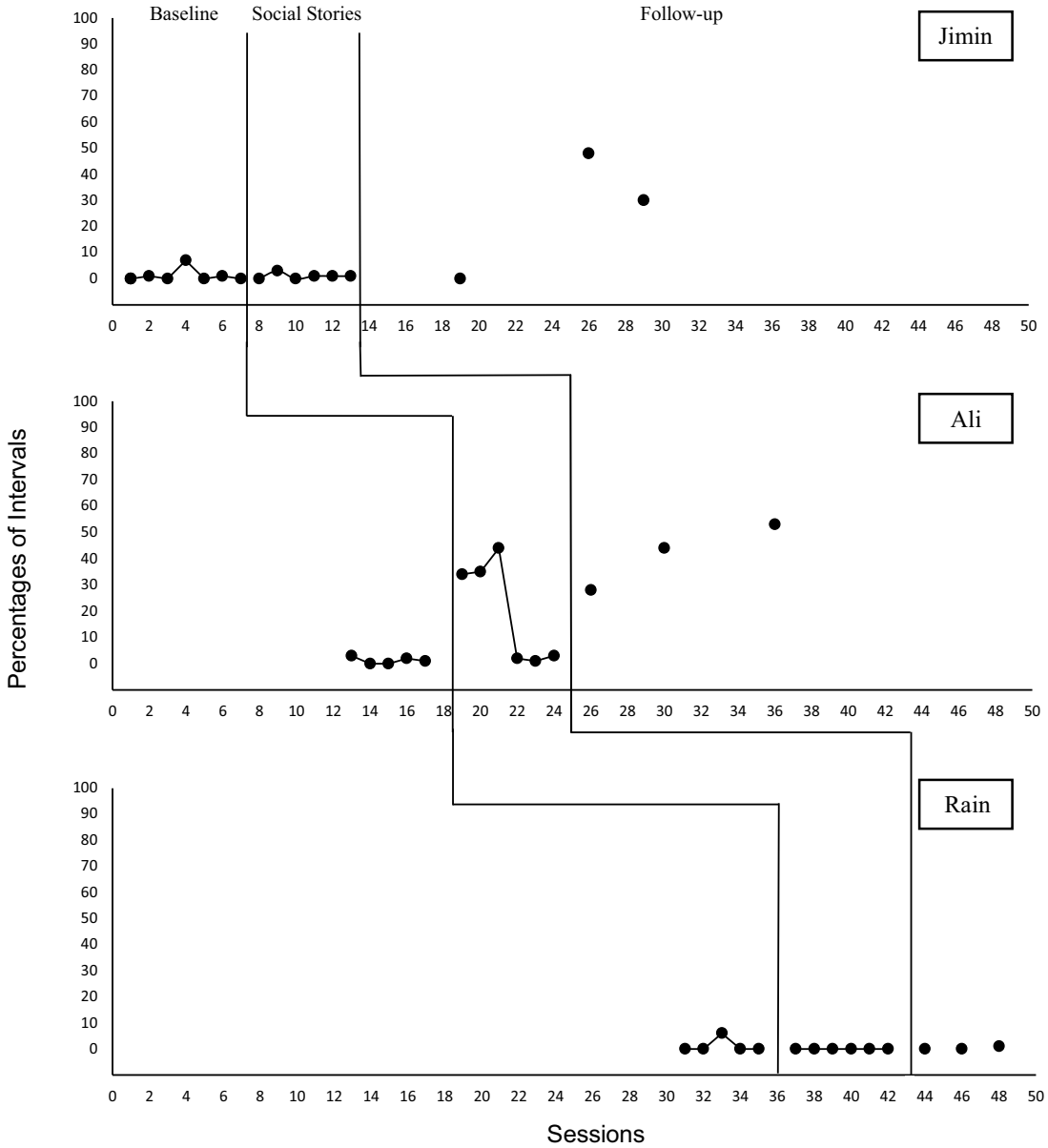


Fig. 3 Estimated duration of social engagement of d/Dhh students with their peers

with their peers, respectively. This duration was estimated by calculating the percentages of intervals of "engaged" interactions during each 20-min session for baseline, intervention, and follow-up phases.

Throughout the baseline condition, all participants who were d/Dhh-ASD spent 0 to 2% of the time engaged with their peers, except for outliers of up to 7%. During the intervention phases, there were little changes in the duration of social engagement of Jimin and Rain, and their peers respectively. Ali and his peers showed an increased level of social engagement for the first half of the intervention phase with a range of 34 to 44%. There was a noticeable decline in the percentage of intervals after the fourth data point and the level remained 1 to 3% of time engaged until the end of the phase.

During the follow-up phase, Jimin did not show any engagement with his peers for the first data point. There was a dramatic increase for the second data point to 48% followed by a decline to 30% engagement for the last data point. Ali and his peers were socially engaged at higher

levels with a range of 28 to 53% (an average of more than 40%), gradually increasing throughout the entire follow up condition. The levels of engagement of Rain and his peers were no different from the previous phases.

Social Validity

Table 3 shows the social validity scores collected from the nine classroom teachers and educational assistants on a Likert-type scale from 1 (strongly disagree) to 5 (strongly agree). According to their evaluation, the average social validity score was 3.92 (ranging from 3.46 to 4.30) during the intervention phase and 4.13 during the follow-up (ranging from 4.03 to 4.02). The average social validity scores for each class were 3.69 (Jimin's), 3.74 (Ali's), and 4.65 (Rain's). Most scores ranged from *neutral* to *agree,* indicating the teachers and educational assistants were likely to agree that the intervention has socially important goals, acceptable procedures, and socially important outcomes.

Discussion

This study evaluated the effectiveness of using social stories combined with teaching ASL vocabulary to increase social interactions and engagement of students who are d/Dhh-ASD and their peers in inclusive school settings. Researchers implemented the intervention in three classes, each of which included a student who is d/Dhh-ASD. The results indicated no functional relation between the intervention and an increase in social interaction across the three students who are d/Dhh-ASD and their peers; rather, the results demonstrated basic effects on all dependent variables for only one student and his peers. These findings differ from previous research in which social stories were identified as an effective strategy for improving social interaction and communication in (a) children with ASD (Chan & O'Reilly, 2008; Crozier & Tincani, 2007; Schneider & Goldstein, 2010), (b) d/Dhh students in inclusive settings (Raver et al., 2014), and (c) classes where students with developmental disabilities were included (Fees et al., 2014). This variability of the effectiveness requires researchers and educators to view the results cautiously before another social story intervention is repeated for students who are d/Dhh-ASD in inclusive settings.

Variable Effects Across Students

As an EBP, social story interventions have been widely used in a variety of settings to support the social development of students with ASD. However, as the guidelines for the implementation have changed over time (Gray, 2004; Gray & Garland, 1993) with little to no research to rationalize the changes (Leaf & Ferguson, 2017; Leaf et al., 2020), some researchers pointed out that the effectiveness of social stories could be variable due to the differences in implementation (i.e., in isolation or in combination with another intervention) and construction (More et al., 2013; Wahman et al., 2019). Thus, the complexity of implementation and construction should be considered carefully for a target population. In this study, considering the unique needs of the students who are d/Dhh-ASD, the intervention was implemented (a) with the whole class, rather than one-on-one, and (b) by modeling ASL vocabulary. The social stories in this study were constructed using the guidelines provided by Gray (2004) but not individualized to each participant and their peers (Fees et al., 2014). These modifications, along with the intervention's own variability in implementation and construction, may be limitations when ascertaining which components of the social story intervention contributed to its variable effectiveness in this study. To interpret the variable effects of the intervention, the following will be discussed: (a) the student skill and language level, (b) delayed effect of the intervention, (c) environmental factors, (d) supporting staff roles, and (e) social validity. Additional modifications (e.g., role play, practice, video models) to the social story interventions and/or additional readings of each story may have been beneficial for all of the students in the class.

Student Skill Level

Although all three participants were in the same grade level, met the inclusionary criteria, and their baseline data points were similar (almost zero to very low), they displayed a wide range of abilities in language and

Table 3 Summary of mean scores of social validity across educators

Statements	Intervention phases				Follow-up phases			
	Jimin	Ali	Rain	Avg	Jimin	Ali	Rain	Avg
Social importance of the goals	3.67	4.33	4.89	4.30	3.33	3.89	4.89	4.03
Acceptability of the procedures	3.89	3.33	4.75	3.99	4.17	4	4.89	4.35
Social importance of the outcomes	3.22	3	4.16	3.46	3.83	3.89	4.33	4.02
Average	3.59	3.55	4.60	3.92	3.78	3.93	4.70	4.13

1 = strongly disagree, 5 = strongly agree

communication, cognitive and physical development, and regulation of behaviors as shown in Table 1. For example, while Ali was able to make requests with three to four ASL signs in a sentence structure, Rain constantly presented requests with physical aggression (e.g., grabbing peers' hair). Although Rain's intention appeared to be genuine engagement or to gain attention with his peers, these types of communicative attempts deterred peers from interacting with him and were not included in the data collection for social initiations or responses. Given the acquisition of successful response and initiation behaviors is considered to be imperative for the establishment and the continuation of successful communication between the student with ASD and their peers (Kok et al., 2002), the three participants were at different readiness levels in their social interactive skills. Some students might need to acquire prerequisite behaviors or need additional modifications to the intervention in order to benefit from social stories (Kalyva & Avramidis, 2005). When examining the efficacy of social stories and considering elemental modifications for the population, the individual differences among students (More et al., 2013; Schneider & Goldstein, 2010) and other features (e.g., partners, physical and social environment; Leaf et al., 2020) should be considered throughout the course of the intervention.

Each of the three participants displayed lower levels of social initiations than responses across the phases, which is consistent with evidence that social initiation skills are more challenging to acquire for most students with ASD (Kabashi & Kaczmarek, 2017; Kalyva & Avramidis, 2005; McDonald & Hemmes, 2003). In contrast, the overall social initiations by peers outnumbered their responses, as presented in Fig. 2. The number of social initiations and responses by peers in each session were higher than those by the student with d/Dhh-ASD in the same class, as with previous studies (e.g., Koegel et al., 2012).

Delayed Effect

There was a delayed effect, as illustrated by the data in the follow-up phases (see Figs. 1, 2, and 3), although the intervention did not appear to have an immediate effect on participants' social interaction. Jimin's responses and his peers' initiations noticeably increased in the follow-up phase. Both Ali and his peers' social initiation and responses comparably improved in the follow-up phase as well. Thus, a longer duration of their social engagement was exhibited.

This delayed effect could be due to one of the unique aspects of the intervention, which involved teaching ASL vocabulary to the whole class. ASL was a new language for most of the peers to learn. It was observed across settings that the peers practiced the ASL vocabulary they learned through the sessions with each other but not as often with the students who are d/Dhh-ASD. Knowledge of ASL by the peers was also not assessed. Participating peers in grade three or four might be shy about using ASL accurately immediately after the intervention, which is common when learning a new language. In this study, each of the six social stories were read only one time to each class over a two-week period, and five different ASL vocabulary were taught at each time, for a total of 30 new ASL vocabulary words. This timeline may not have been sufficient to investigate the effectiveness of this intervention, considering the time it takes to acquire a language. Future research should examine the determination of adequate dosage (i.e., number of times or frequency of dose; Parker-McGowan et al., 2014), the number of words taught per session and the total number, an evaluation of acquisition of the ASL vocabulary, and the duration of each social story intervention (Schneider & Goldstein, 2010) in order to generalize the delayed effect.

Environmental Factors

Some environmental factors could support the delayed effect during the follow-up phase, which was quite evident in Jimin's and Ali's classes, contrary to Rain's. Rain was the only deaf student in his school, while Jimin and Ali were not. Jimin's and Ali's schools also had staff who were d/Dhh. There were also school-wide events (e.g., holiday concert with the school for the deaf) or support programs (e.g., ASL club) in Jimin's and Ali's schools where their peers were continuously exposed to ASL. The school's social environment may have supported Jimin's and Ali's peers' use of ASL with their d/Dhh-ASD peers moreso than Rain's.

The classrooms' physical environment might have impacted peer engagement (Giangreco et al., 2010; Guardino & Antia, 2012). All three classes had the same seating arrangement: four to five desks were physically touching in pods. While Jimin and Ali were seated in one of the clusters, Rain was only paired with his EA at the back of the classroom. This physical environment likely limited Rain's social engagement with the rest of his class and influenced his behaviour as well as the behaviour of the other students towards Rain (Giangreco et al., 2010).

Supporting Staff

As a possible confounding variable, the one-on-one EAs' previous experiences, knowledge, and skills may have affected the intervention. Although all three students who are d/Dhh-ASD received full-time EA support, each EA's skill sets and roles were quite different. Jimin's EA had previously worked as a behavioral interventionist for children with ASD; therefore, he may have a deeper understanding of effective social skill instruction. It was observed during

the follow-up phase that he facilitated social interactions between peers through coaching in the classroom and on the playground. In Ali's class, the EA supervised Ali very closely and continued conversations with him during the observations for the second half of the intervention phase. During this time his peers rarely approached Ali, which led to a drop in data points in the middle of the intervention. During the follow-up phase, the school removed the supervision, and some peers invited Ali to play with them and were socially engaged with him for most of recess. For Rain, a new EA was assigned to him on the day the intervention phase started. Although the EA was proficient in ASL skills, it was her first EA position, and Rain's behavior issues escalated while adjusting to working with a new person. A part of the role of EAs who work with students who are d/Dhh-ASD in inclusive settings is to facilitate peer social interactions. Thus, in-service training for support staff should be provided to increase daily instances of positive interaction through general instruction regarding social skills and specific teaching of playground games and rules (Vicker, 1998; Wahman et al., 2019).

Social Validity

The average in the follow-up phase (4.13) increased across all three classes compared to the average in the intervention phase (3.92). This may support the delayed uptake of the intervention by the educators since the student participants showed delayed effects as well. Therefore, the period of intervention may need to be extended to see more long-term effects on social interaction and communication.

As shown in Table 3, Rain's classroom teacher and EAs rated social validity as the highest across all three classes in all three areas: goals, procedures, and outcomes of the intervention (4.89, 4.82, and 4.25 respectively) although Rain's class showed no progress. There were no interviews as to why they perceived the social validity of the intervention so highly, but two possibilities could be speculated. First, compared to the other two classes, there had been little support for Rain and his peers to socially engage with each other and receive support from a behavioral analyst or the benefits of a school-wide inclusion program. Prior to the study, Rain's teachers shared their struggles with finding ways to improve social inclusion in his class and appeared to look forward to having the opportunity to be part of this study. Second, the educators reported that when the social stories and ASL vocabulary were introduced to the students, the class's overall engagement throughout the day increased. Students approached the researcher to check if their signs were correct and share their experiences of using ASL with Rain. Rain's classroom teacher also stated how much her class enjoyed the intervention and asked for extra materials to help them continue learning ASL after the study was completed. Although not reflected in the single case design data, the social validity scores provide a meaningful measure to show the influence of the intervention to the participant's class.

Limitations

The limitations of this study include generalizability and inability to isolate the effect of including ASL vocabulary in teaching the social stories. This study was conducted at three schools with only three students who are d/Dhh-ASD. Future research is needed to determine if the results could be generalized to a greater population of students and in different learning environments. As previously noted, there were possible confounding variables that may have contributed to variability in intervention effects (e.g., EA skills, ASL club). A replication study could be designed to control for these variables and determine a more accurate effect of the intervention. Researchers should also attend to the contributions of each intervention component to determine whether behavioral changes are the result of social stories, the impact of the other component (i.e., ASL vocabulary teaching), or a combination of both (Schneider & Goldstein, 2010; Wahman et al., 2019).

Another limitation, or as a possible reason for the lack of treatment effect to increase social interactions, might be related to the time of observation. As previous research recommended to observe social interactions during unstructured, non-instructional time (Kurth et al., 2015; More et al., 2013; Owen-Deschryver et al., 2008), data collection in this study took place at lunch and recess time. However, 86% of all interactions occurred during recess and only 14% during lunchtime. This may be due to the classroom lunch environment, as students stayed at their desks while having lunch and most of the students hurried to finish their lunch to go outside for recess. Some peers had conversations naturally at their clustered desks while eating; however, some participants who are d/Dhh-ASD were not able to access their peers'conversation due to limited auditory input or their peers' lack of ASL skills. In future research, consideration of the times when students have the most opportunities for social interaction could be individualized for each setting/school environment.

Implications for Researchers

There is a great need to modify interventions that have been found to be effective in adjacent fields for individuals who are d/Dhh-ASD (Borders et al., 2022; Bruce & Borders, 2015). Social story interventions may be utilized in combination with other developmentally appropriate strategies or interventions depending on a student's abilities and

situations (Leaf et al., 2020; Vicker, 1998; Wahman et al., 2019). For example, when a student who is d/Dhh-ASD experiences isolation in the classroom and develops inappropriate communicative engagement, a social story intervention might not be effective by itself to increase social interactions with peers. Rather, it might have a more powerful impact when combined with other tailored strategies (e.g., functional communication training, role modeling, video modeling) as part of their positive behavior support program. As most of the peer initiations were made by two or three of the same peers, the researchers suggest incorporating a peer-mediated training (e.g. Circle of Friends) or direct coaching approach with volunteering peers. Increasing social interactions require reciprocal efforts, not only by the targeted student in the inclusive environment but by the whole class community.

Conclusion

Social interactions between students who are d/Dhh-ASD and their peers are essential for success in inclusive educational settings and can improve the quality of life for individuals with disabilities. In order to best meet the complex needs of this population, identification of effective interventions is necessary. The current study investigated an intervention that combined teaching social stories and ASL vocabulary. While the results may not indicate a clear treatment effect across the three participants and their peers, it is important to note the study is the first social story intervention to be conducted with students who are d/Dhh-ASD and their peers as a whole class. Possible confounding variables include individual differences and environmental factors, which may explain the various effects of the intervention. In addition, students might need additional supports or practice to improve social communication and proficiency with ASL, including in vivo rehearsal or coaching, as well as peer-mediated instruction.

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Declarations

Conflict of interest We have no known conflict of interest to disclose.

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